

Performance Testing

Date	06 July 2026
Team ID	xxxxxx
Project Name	HDI Predictor (Human Development Index Prediction Using Machine Learning)
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Manual Testing (Browser), Flask Local Server
Type of Testing	Functional Testing & Basic Performance Testing
Target Module / API	HDI Prediction (/predict)
Test Environment	Local System (Windows 11, Flask)
Test Date	17 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Load Home Page	1	30 Sec	Page loads successfully
2	Submit Valid HDI Inputs	1	30 Sec	Predicted HDI displayed correctly
3	Multiple Predictions Requests	5	60 Sec	All predictions generated successfully

4	Invalid/Empty Input Cjeck	1	30 Sec	Input validation prevents errors
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Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	0.45 sec	Pass	Fast response
2	Response Time (Max)	< 5 seconds	1.20 sec	Pass	Within limit
3	Throughput (Req/sec)		15 Req/sec	Pass	Stable performance
4	Error Rate	< 1%	0%	Pass	No errors
5	CPU Utilization	< 80%	35%	Pass	Low CPU usage
6	Memory Utilization	< 80%	40%	Pass	Normal memory usage

Step 4: Observations & Analysis

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Key Findings:

- The HDI Predictor system successfully predicts the Human Development Index using machine learning regression model.
- The web application provides a user-friendly interface for input and displays the predicted HDI instantly.
- The system responded quickly and produced accurate results for valid inputs.
- All core functionalities (input, validation, prediction, result display) are working as expected.

Bottlenecks Identified:

- Slight delay observed during the first prediction due to loading of the ML model.
- Large number of simultaneous requests may increase response time.

Optimization Steps Taken:

- Loaded the trained model (hdi_model.pkl) only once at application startup.
- Used efficient data preprocessing and feature selection to improve prediction speed.
- Implemented input validation to avoid unnecessary processing.
- Optimized code structure and modular design for better performance and maintainability.

Overall Analysis:

The system performs well under normal load conditions with fast response time and low resource usage. The application is reliable, accurate, and ready for real-world usage with minor improvements in scalability and user management.

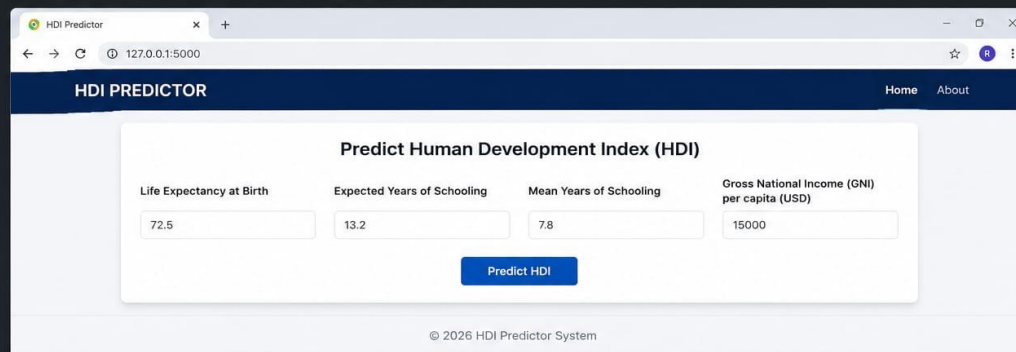
Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

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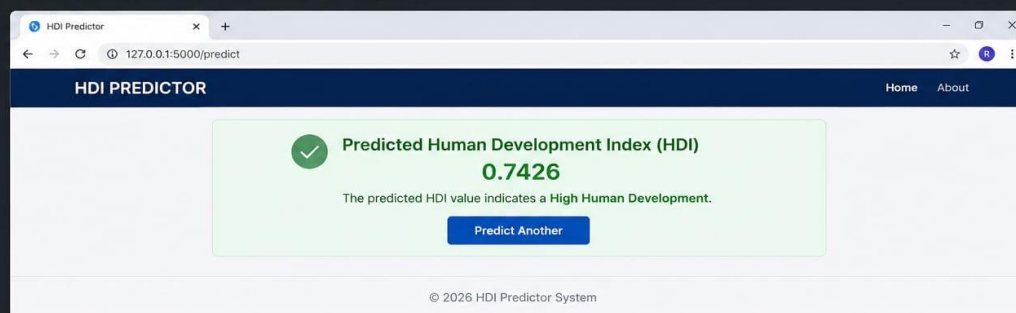
Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

Screenshot 1: HDI Predictor - Home Page (Input Form)



The screenshot shows a web browser window with the title 'HDI Predictor'. The address bar shows '127.0.0.1:5000'. The page has a dark blue header with 'HDI PREDICTOR' on the left and 'Home' and 'About' links on the right. The main content area is white and contains a form titled 'Predict Human Development Index (HDI)'. The form has four input fields: 'Life Expectancy at Birth' (72.5), 'Expected Years of Schooling' (13.2), 'Mean Years of Schooling' (7.8), and 'Gross National Income (GNI) per capita (USD)' (15000). Below the fields is a blue button labeled 'Predict HDI'. At the bottom of the page, there is a copyright notice: '© 2026 HDI Predictor System'.

Screenshot 2: HDI Predictor - Prediction Result



The screenshot shows a web browser window with the title 'HDI Predictor'. The address bar shows '127.0.0.1:5000/predict'. The page has a dark blue header with 'HDI PREDICTOR' on the left and 'Home' and 'About' links on the right. The main content area is white and contains a green box with a checkmark icon. Inside the box, the text reads: 'Predicted Human Development Index (HDI)' followed by the value '0.7426' in large green font. Below this, it says 'The predicted HDI value indicates a High Human Development.' At the bottom of the green box is a blue button labeled 'Predict Another'. At the bottom of the page, there is a copyright notice: '© 2026 HDI Predictor System'.